

Enterprise large language models: Knowledge characteristics, risks, and organizational activities

Daniel E. O'Leary

Marshall School of Business, University of Southern California, Los Angeles, California, USA

Correspondence

Daniel E. O'Leary, Marshall School of Business, University of Southern California, Los Angeles, CA, USA.

Email: oleary@usc.edu

Summary

Since the release of OpenAI's ChatGPT, there has been substantial interest in and concern about generative AI systems. This paper investigates some of the characteristics, risks, and limitations with the enterprise use of enterprise large language models. In so doing, we study the organizational impact, continuing a long line of research on that topic. This paper examines the impact on expertise, the organizational implications of multiple correlated but different responses to the same query, the potential concerns associated with sensitive information and intellectual property, and some applications that likely would not be appropriate for large language models. We also investigate the possibility of agents potentially manipulating the content in these large language models for their own benefit. Finally, we investigate the emerging phenomenon of "ChatBot Enterprise" versions, including some of the implications and concerns of such enterprise large language models.

KEYWORDS

BARD, BlenderBot, ChatGPT, ChatGPT Business, ChatGPT Enterprise, enterprise generative AI, enterprise large language model (ELLM), experimental analysis, expertise, generative AI manipulation, generative AI systems (GAIS), human-in-the-loop, large language models (LLM)

1 | INTRODUCTION

Recently, McKinsey¹ indicated that "Generative AI has the potential to change the anatomy of work" Forbes² asked "Will AI Take Over The World?" As a result, it seems as if we have seen many people throw their hands up, saying that generative artificial intelligence (AI) will take over enterprises, that there will be far-reaching job losses, and wondering if there would be anything for people to do besides ask ChatGPT. Maybe the world will change; however, in this paper, I try to take an alternative approach, analyzing some current characteristics of generative AI knowledge and investigating its use

for an enterprise environment. In so doing, I address issues such as the impact of the similarity of generative AI systems (GAIS)/large language models (LLM) responses to the same queries, examine the role of the human-in-the-loop (HITL) use of these systems, analyze the potential impact of using such systems when there is sensitive information or intellectual property, and lay out some enterprise activities that do not seem to fit well with generative AI capabilities. I also analyze the potential for manipulation of LLM by agents other than the developers. Finally, I examine the recent phenomena of "Enterprise ChatBots," such as ChatGPT Enterprise or ChatGPT Business, where secure versions are brought into enterprises and managed for the organization's own use.

One view of LLM is that they are another form of knowledge that can be used as part of a portfolio of knowledge management (O'Leary, 2000). Accordingly, we summarize some of the characteristics of generative AI knowledge and some of the risks. Further, as far back as O'Leary and Turban (1987), Duchessi et al. (1993), and O'Leary and O'Keefe (1997), researchers have been investigating the

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¹<https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/the-economic-potential-of-generative-ai-the-next-productivity-frontier>

²<https://www.forbes.com/sites/forbesbooksauthors/2023/07/17/will-ai-take-over-the-world-or-will-you-take-charge-of-your-world/>

organizational impact of AI on enterprises. This paper continues that analysis by focusing on the impact of LLM in some organization settings.

1.1 | Sources of knowledge

LLM draw from a broad range of text sources. Schaul et al. (2023) analyze the websites from which these LLM have been built. They indicate that there is information from over 15 million websites in the systems but that roughly 5 million of those links no longer exist. Among the leading websites that have been used were [patents.google.com](#), [Wikipedia.org](#), [scribd.com](#), [nytimes.com](#), [journals.plos.org](#), and [latimes.org](#). Accordingly, large chunks of human knowledge are accounted for from these sources. However, those lists also provide insight into where and what knowledge is not accounted for. As noted by Schaul et al. (2023), there is a bias toward western religious sites. Further, larger English-speaking newspapers are represented, likely leaving local information gaps from other parts of the United States and the world. In addition, there likely are domains that are less represented than others. In addition, issues such as dialectal differences are ignored (Sun et al., 2022). Finally, because the information in a LLM has been gathered from the Internet world, information that does not appear in digital form or on the Internet is not directly accessible for inclusion. Accordingly, much intellectual property, secret information, or undisclosed information is not included; however, apparently much copyrighted material is embedded in these LLM.

1.2 | Purpose and contributions

The purpose of this paper is to investigate some of the issues associated with the organizational impact of LLM. As part of that analysis, this paper has several contributions. First, the analysis of LLM suggests that the knowledge in LLM is easy to access and broadly available, so we ask, “does knowledge-based competitive advantage still exist, and if so, where does it come from?” Second, we investigate responses by ChatGPT to the same query and find that the responses are correlated but different. As a result, we ask, “what is the impact of those differences in enterprise settings?” Third, we analyze the potential impact of using LLM in the enterprise, with respect to sensitive information or intellectual property, and ask the question, “what happens when enterprises put information in LLM?” Fourth, we investigate some applications where the use of enterprise LLM is unlikely to be appropriate. Fifth, we ask how can these large language systems be manipulated by external agents. Finally, we investigate the very recent phenomena of “Enterprise ChatGPT,” as firms begin to explore bringing these systems in-house.

This paper does not suggest that we put “guard rails on AI” or “police AI” or even limit development of enterprise AI applications and uses. Even if we could find ways to enforce or accomplish those

tasks, I am sure that people would find a way around those guard rails. However, enterprises and other organizations, need to be aware of the risks of using such systems. Throughout, it is assumed that enterprises see enterprise LLM as an opportunity to use in the support of a broad range of applications. However, we do identify some potential risks that enterprises should consider policies to provide some controls over those risks and identify applications that might not work as well as others.

1.3 | Plan of this paper

This paper proceeds as follows. Section 2 examines some characteristics of generative AI knowledge, while Section 3 examines some of the risks of LLM. Section 4 examines the use of experiments to analyze the extent to which these systems generate different but correlated responses. Section 5 briefly reviews the organizational impact of those correlated but different solutions on the HITL. Section 6 investigates the potential effects of inputting sensitive information into LLM. Section 7 examines the potential use of generative AI in some enterprise tasks. Section 8 investigates an alternative concern, investigating how might enterprises manipulate and influence the content of a LLM? Section 9 investigates some of the implications and concerns of bringing chatbots into the enterprise. Finally, Section 10 summarizes the paper, examines its contributions, and provides an extension to this discussion.

2 | CHARACTERISTICS OF GENERATIVE AI KNOWLEDGE/LARGE LANGUAGE MODELS

This paper reviews some of the characteristics of LLM that could influence the affects that those systems have on organizations. Those characteristics include that the knowledge in these systems is inevitably historical in nature, it is easily and broadly available, making the knowledge that they contain a commodity of sorts, ultimately influencing what would likely be “expertise.”

2.1 | Historical knowledge

To facilitate broad use of the results of generative AI, LLM are used to generate new neural network updates only periodically. As an example, ChatGPT will tell you that it's last “knowledge update” was as of September 2021. As a result, these systems are built on an “historical” perspective or view of the world. Unfortunately, GAIS face the problem of being “so yesterday.”

However, LLM can be integrated with search engines to mitigate some of those concerns of being outdated. For example, Microsoft's Bing gives up-to-date search information, while BARD often includes links in its answers. Further, these Chatbots can be integrated with other knowledge bases and applications.

2.2 | Ease of access to knowledge

LLM continue the development of the digital world, providing access to a representation of much of that knowledge. Virtually anyone with access to a computer and the Internet can access LLM. In addition, LLM speak multiple languages. A frequently asked questions (FAQ) for BARD indicates that it speaks more than 40 languages. As a result, this means that access to LLM expertise is available up and down the organization and around the world, internationally.

2.3 | Knowledge as a commodity

Because of that broad and easy access, Bard, ChatGPT, and others offer the same capabilities to virtually all potential users. As a result, at one level, their use and the information that they provide are a commodity. Perhaps, by being clever, people can generate something new and different, but the base capability is offered to all. Competitors have access to LLM. In an organizational hierarchy, supervisor and employee have access to the same information. Around the world, all have access to the same knowledge. Accordingly, a key question to ask is “if we use the same information as other organizations, what will be the impact on our stakeholders?”

2.4 | Impact on expertise—Knowledge-based competitive advantage

If all have access to the same knowledge that is broadly available, where does knowledge-based competitive advantage come from? Competitive advantage would come from the unique use of such tools, other sources of knowledge, or from alternative sources of expertise not captured or embedded in LLM.

When using such systems, it is likely that insightful question asking expertise and curiosity would provide the best insights from LLM. However, that is an empirical issue and potentially could be addressed by some experiments with GAIS.

What might be other sources of knowledge? As noted above, although there may be information from over 15 million websites (Schaul et al., 2023), inevitably, there will be limitations and gaps in the knowledge. As an example, O'Leary (2023a) noted that ChatGPT could not correctly identify the originators of case-based reasoning. Unfortunately, based on that research and queries by researchers, it is clear that the systems may not “know” the gaps in their knowledge and do not “know” that they do not know the gaps. Further, such systems do not have access to enterprise knowledge management systems and internal knowledge.

This discussion leads to the question “What kinds of expertise will be the most valuable in a LLM world?” It is likely that “recent” knowledge and expertise is more valuable than historic expertise, because a LLM is largely historical and only as up-to-date as of its latest version. In addition, unstructured knowledge that may not be well-captured in text would also be more valuable than structured

knowledge that has been captured in text and is reproducible. Perhaps, such unstructured knowledge will derive from individual and organizational experience or other factors. Finally, because these LLM do not have access to internal corporate knowledge, experts on internal processes, activities, and so forth provide valuable expertise beyond their direct knowledge.

Because GAIS knowledge is historical and all have access to it, then perhaps some internal knowledge, human expertise, and unstructured knowledge have never been more valuable? But with access to such broad knowledge in LLM, it is likely that it will become more difficult to become an expert of note over time.

3 | SOME RISKS OF LLM

There are a number of risks associated with the use of LLM, including accuracy, reproducibility, bias, lack of explanation, security concerns, and other risks. Issues of accuracy have been discussed by many including O'Leary (2015, 2019, 2022, 2023a) and Neumeister (2023). O'Leary (2023a) noted LLM did not seem to know much about “who” the key developers were of different innovations in AI. Neumeister (2023) reviewed a case where a lawyer used ChatGPT to generate what ultimately was determined to be “fictitious legal research,” including cases that were “fake opinions.”

O'Leary (2023b) found that both ChatGPT and Bard did not provide the same answer to multiple identical queries. In his research, O'Leary (2023b) found that the systems provided different answers to the same query when using questions from the television game, Jeopardy! In that research, he also found that not only were the answers different but they were sometimes incorrect. Bias has long been a recognized issue and seen as a danger in the use of such systems. As an example, Vincent (2016) discussed how a Microsoft effort had resulted in racist and other concerns. More recently, DeGeurin (2022) investigated inappropriate behavior in BlenderBot. Researchers have noted that the root of such behavior is in the text on which they are based. Questionable language has been a problem with intelligent systems. As an example, IBM developers found inappropriate language when they input the “Urban Dictionary” into Watson, ultimately forcing them to pull the dictionary from Watson's knowledge base (Madrígal, 2013).

LLM are developed using neural nets, which are well known for providing answers without explanation. The data-driven, black box approach limits the ability to generate an explanation from these systems. In addition, the amount of text used by these systems is so vast and the development teams are so large that not even the developers know the origins of specific responses and it is difficult to track such origins.

Unfortunately, the ability of LLM models to generate human sounding text suggests that these systems can be used to create fake news or fake images and so forth. Accordingly, these systems can be “weaponized.” As an example,³ Binance recently indicated that ChatGPT was used “... to falsely claim it has Communist ties.”

³<https://protos.com/binance-says-chatgpt-was-weaponized-to-falsely-claim-it-has-communist-ties/>

Building LLM are very costly and require substantial resources. As an example, Narayanan et al. (2021) trained the GPT-3 model on 300 billion tokens for 34 days, using 1024 A 100 graphics processing units (GPUs). As a result, the power requirements alone raise questions of sustainability risks with the development of these systems, particularly in light of the mushrooming numbers of systems being developed.

4 | HOW SIMILAR ARE LLM RESPONSES TO THE SAME INPUTS?

An important question in the use of LLM is “how similar are its responses to the same questions and inputs?” If each query generates the same response, then there is the equivalent of a “single truth.” If there is a single truth, then each user will see the same information with the same query. This issue is an empirical question, so one approach to better understand the various behaviors and characteristics of LLM is to run experiments to address this question. In this section, we briefly review some experiments that provide insight into the “uniqueness” of the information that they generate for users. Although the experiments were limited, the results indicate that responses from the system are likely to be correlated but a bit different.

4.1 | Uniqueness of responses to a question

O'Leary (2023a) provided a question to ChatGPT about “innovative organizations” and regenerated three additional versions to the question. Those responses were then sequentially placed into Turnitin, software that compares the content to other content in its database. That test found that the four essays were similar at the four different percentages: 0%, 32%, 42%, and 61%. The second essay was similar to the first, the third to the second and third, and the third to each of the other three. These results suggest that if different users input a question into ChatGPT, then those results are likely to be similar, and correlated with each other, but also a bit different.

4.2 | Uniqueness of reviews of the same paper

In a similar test, I inputted an essay into ChatGPT three different times and asked the system to “review” that essay. After putting the paper into the system, I hit “regenerate” two different times. The results are summarized in Table 1. Unlike the results above, we did not find a monotonic increase in the measured similarity. The similarity between the nine different reviews ranged from 0% to 28%, but again, the results suggest that the answers were correlated with each other but also were a bit different.

TABLE 1 Turnitin similarity of system-generated reviews.

Number	Similarity	Task
Set 1-1	0.04	Entered text
Set 1-2	0.11	Regenerate
Set 1-3	0.28	Regenerate
Set 2-1	0.28	Entered text
Set 2-2	0	Regenerate
Set 2-3	0.23	Regenerate
Set 3-1	0.14	Entered text
Set 3-2	0.12	Regenerate
Set 3-3	0.11	Regenerate

5 | HUMAN-IN-THE-LOOP

LLM require a HITL. A human will use the system to talk with, to ask it questions, and to have it do things. However, as noted in O'Leary (2003), given the same information, different HITLs interpret the same information differently. This could lead to different decisions and actions from the HITL for the exact same query responses.

In accounting, finance, and other organizational settings, enterprises talk of a “single source of the truth.” There is a desire to have decision-makers all see the same information, so that their decisions are aligned and that the organization can move forward without being drawn in different directions. However, as noted above, LLM can provide multiple correlated versions of the truth.

As a result, unfortunately, neither the human use nor the LLM response is totally identical from use to use. Thus, the LLM and the HITL put multiple layers of uncertainty into the use of these systems, potentially making it difficult to “align” decision-making in an organization.

6 | SENSITIVE INFORMATION/ INTELLECTUAL PROPERTY

For enterprises, a key risk always is the potential disclosure of sensitive information or intellectual property to the world. As a result, this raises the question, “if private intellectual property is disclosed to the system, will the system diffuse that information to others?”

Unfortunately, as noted by Gartner (2023):

There are currently no verifiable data governance and protection assurances regarding confidential enterprise information. Users should assume that any data or queries they enter into the ChatGPT and its competitors will become public information, and we advise enterprises to put in place controls to avoid inadvertently exposing IP.

An analysis of OpenAI's privacy statement⁴ states that "Network Activity Information, such as Content and how you interact with our Services" can be disclosed to "... our affiliates, vendors and service providers, law enforcement, and parties involved in Transactions." Of course, this indicates that the information is available, but not that it is used or not used.

As another example, based on BARD's policy statement, it appears that Google does collect information about conversations. "When you interact with Bard, Google collects your conversations, your location, your feedback, and usage information. That data helps us provide, improve and develop Google products, services, and machine-learning technologies"⁵

This finding that information from conversations may be gathered probably is not surprising, because there are a number of reasons for those firms developing LLM to want to gather such conversations. From the perspective of improving the system, LLM providers would be interested in gathering content, particularly in those settings where the user indicated a problem or indicated that they were not pleased with the system's response. It is also easy to understand that law enforcement might want information about conversation content.

As a result of these considerations, a key development in most organizations would be the development of appropriate policies to limit employees from placing inappropriate information and intellectual property into LLM.

7 | SOME POTENTIAL ENTERPRISE ACTIVITIES

LLM offer the ability to perform several important enterprise activities. For example, as illustrated with Microsoft's Bing, LLM can help users shop and LLM can answer questions that historically would have required a human. As a result, many such customer interaction activities would seem to be potentially become part of digital transformation. However, our analysis leads us to some activities that we expect should not use LLM.

7.1 | Responding to proposals

A common business situation is when an organization issues a proposal for goods or services. There are at least three concerns associated with using a LLM to help generate a response to that proposal. First, our results on the similarity of LLM responses suggest that vendors responding to a proposal by using LLM likely will generate similar or correlated proposals. If the results are similar, then that may suggest collusion, when in fact, there had not been any specific communication between the vendors. Suspicion of collusion would have many negative reputation effects. Second, if vendors do use GAIS, then there is the well-known potential for system hallucination. For

example, in law, a brief was filed with a court that included some cases that were not real. ChatGPT apparently had made up the cases (Neumeister, 2023). Accordingly, proposal issuing companies may require as part of response to a call for papers, an attestation to the client company that either the response was written by a person or that the signee has read the response and takes responsibility. Third, if the response to the proposal is considered proprietary, then, as noted above, information used with the LLM may become public information, leaking valuable intellectual property to competitors.

7.2 | Competitive intelligence: Use an LLM to anticipate a competitor's strategy?

A key issue in competitive analysis is the investigation of competitors' strategies. Unfortunately, because of the historical nature of knowledge in LLM, it does not seem that LLM will be an effective tool to study the strategies of other companies. As an example, I asked:

Me: "What is KPMG's audit strategy?"

ChatGPT: "As of my last knowledge update in September 2021, KPMG is one of the 'Big Four' However, I do not have access to specific, up-to-date information on their current audit strategy beyond that point. ... It's important to note that audit strategies can evolve over time in response to changes in regulations, technological advancements, and shifts in the business landscape. To get the most accurate and up-to-date information about KPMG's current audit strategy, I recommend visiting their official website or referring to their recent publications."

The historical nature of LLM may limit their use for tasks for which up-to-date information is required. In addition, tasks, such as competitive intelligence, likely would involve gathering any nondigital and secret information that can be captured and used.

8 | MANIPULATING AND INFLUENCING THE CONTENT OF GAIS/LLM

Although this paper has focused on the risks and impacts of these LLM on enterprises, there also is a related risk facing developers: Can a person or enterprises directly affect the content of the LLM? If so, then how can these systems be influenced by external parties?

If someone is interested in getting information into these LLM for their own or for enterprise purposes, one approach would be to provide the information to the sites identified as being sources of knowledge, such as those identified in Schaul et al. (2023), and then maintain that information, as these systems refresh their knowledge over time. For example, a Wikipedia page could be developed, or information could be provided to personal blog sites. Alternatively,

⁴<https://openai.com/policies/privacy-policy>

⁵<https://bard.google.com/faq>

because these systems use information gathered from websites, perhaps, having a sequence of websites would be another approach. In this manner, information could be provided to firms developing these systems to influence the results that the systems generate.

Another approach may come from the introduction of information into these systems. As noted above, if conversations are kept and used, then a way to influence the content of the systems, perhaps for your own purposes, is to put the information that the agent wishes to diffuse, into the model. It is easy to imagine a room full of spies busy putting false information into a system in order to change its knowledge.

As an example, authors may want their research recognized by these systems. Perhaps, by building history pages or maintaining pages in Wikipedia or having web pages, information about those researchers may be captured and used by the system. As a result, ultimately, building these systems is likely to require substantial efforts to fend off a broad range of manipulation attempts, while still including useful information.

9 | ENTERPRISE LARGE LANGUAGE MODELS

Perhaps, because of these many limitations of large language systems for enterprise use, recently, OpenAI announced “ChatGPT Enterprise, an “enterprise large language model“ chatbot designed to let enterprises “control their own data” (Field, 2023; OpenAI, 2023). In ChatGPT Enterprise, OpenAI does not train on the business's data, organizations can have their own data retention policies, and the enterprise owns the inputs and output. Of course, this technology has several implications, including the need to generate the organization structure, policies, and personnel necessary to manage this new source of knowledge. In addition, as above, organizations need to embed this technology in their existing systems and strategies and understand where competitive advantage can come from.

For those organizations that choose this approach to knowledge management, there will be some critical decisions to make that likely will require substantial infrastructure. First, firms will need to determine and implement data access policies. If the system is used to generate intellectual property, then it will be important to determine internally who should be able to access that information and how many distinct versions of the system should be in play. Second, if firms control their own data, then they will need to determine the life cycle of the data in the system. For example, if a new marketing campaign is initiated, the system will likely bias responses to knowledge of previous marketing campaigns. Third, because use is likely to influence the system's behavior, organizations will need to generate use policies; otherwise, the system will learn inappropriate information. As noted earlier, the introduction of the “Urban Dictionary,” into Watson, ultimately required removal, because of the system's behavioral change. It is easy to imagine how a variety of enterprise users could gradually make an enterprise version useless. Fourth, knowledgeable leaders will be needed to help guide the use of such

systems. Fifth, continued experimentation is likely to be important to understand how these evolving knowledge sources can be used and how they evolve. Sixth, it is likely that these systems will be integrated with other enterprise systems and knowledge management tools and potentially integrated into workflows. Such integration can facilitate customer service and internal system use through integration with customer relationship management, enterprise resource planning systems, and other systems.

There are likely to be some limitations of using such systems internally. As noted earlier, oftentimes, firms are concerned with ensuring that there is a single truth in the data used in the corporation, so that decision-makers are using the same data and viewpoint. Unfortunately, as noted in O'Leary (2023b), these systems do not necessarily provide the same answer to the same question in each use. Further, as noted above, if the versions do learn based on system use, then potentially, the content can be manipulated by internal users, perhaps for their own benefit. As a result, firms potentially will need to monitor use. In addition, because multiple solutions are correlated, perhaps, the use of such systems will create more “groupthink” within organizations as the same resource is consulted throughout the organization, and correlated solutions are generated.

10 | SUMMARY, CONTRIBUTIONS, AND AN EXTENSION

It is clear that the large language model/generative AI world is moving very fast, and any of those changes could have an impact on organizations and their use of these systems. The results of any analysis or comparison of systems could change with a new system versions and capabilities. As an example, Bing released a modification that allows the user to choose between different types of conversations: creative, balanced, or precise. However, in this paper, I have tried to identify some issues related to the impact of GAIS/LLM on organizations.

10.1 | Contributions

This paper has several unique contributions. First, our analysis of LLM systems leads to note that their knowledge is historical, easily accessible, and a commodity, leading us to conclude that the “right” human expertise has never been more valuable but that becoming an expert has never been more difficult. Second, LLM will produce correlated but somewhat different responses to the same questions, having implications for the use of such systems in and across different organizations. Third, because LLM have an HITL, if organizations use such systems, they may find that the organization decision-making is not in alignment, because of different system responses and different people using the same information differently. Fourth, there is evidence that the use of LLM can lead to private information becoming public information; as a result, organizations need to establish governance limiting the nature of the use of such systems. Fifth, certain organization activities likely should not use LLM, such as responding to proposals,

because of the potential to generate correlated solutions to the same inquiries. Sixth, we identify a risk for developers of these systems that may find agents generating content to manipulate the results in these LLM. Finally, we investigate some potential issues associated with enterprise large language systems.

10.2 | Extension

There are several potential extensions to the paper. One of the more interesting relates to the notion that although current LLM are based on roughly 15 million websites, of which 5 million are no longer available. As a result, if there is a new web scrapping effort, it would not include the knowledge from those 5 million sites, and unless the new efforts include all of the old information, then that information will not be available to the systems. This would raise a concern about the extent that shifting knowledge from those 5 million websites would change the behavior of the systems. In addition, it likely would be considered a risk that one third of websites in such systems could disappear in such a short time.

DATA AVAILABILITY STATEMENT

Data is available from the author.

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